SMART INDIA HACKATHON 2025

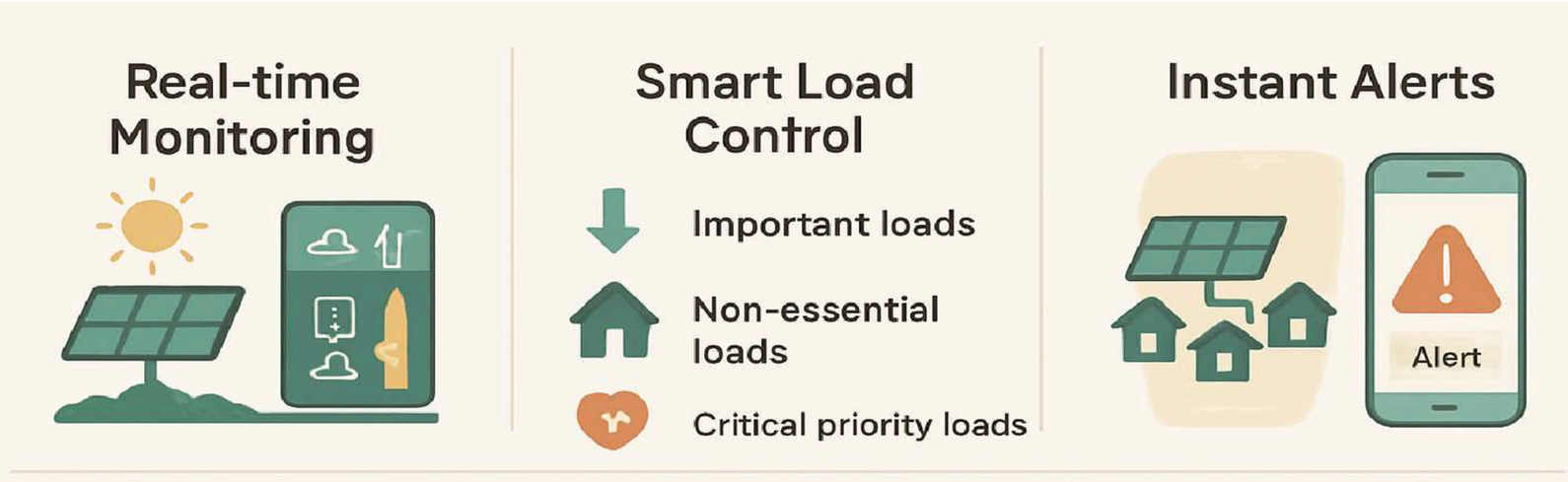
TITLE PAGE

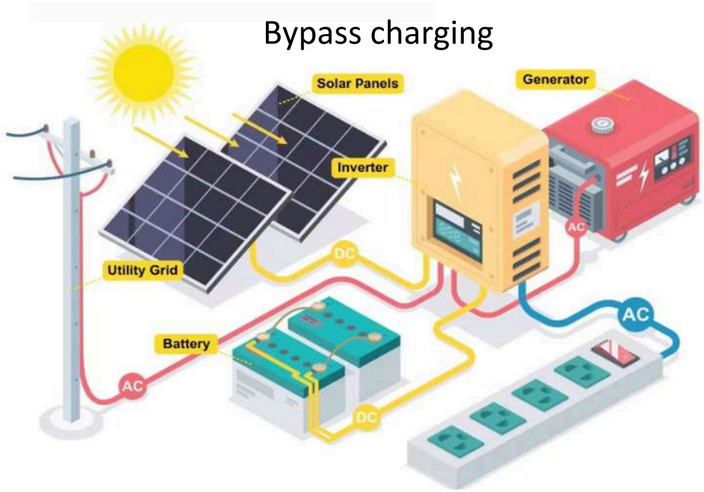
- Problem Statement ID – SIH25051
- Problem Statement Title - Renewable Energy Monitoring System for Microgrids
- Theme- RENEWABLE/SUSTAINABLE ENERGY
- PS Category- Hardware
- Team ID-
- Team Name (Registered on portal)- EcoFlux

IDEA TITLE

The primary reason is the lack of proper monitoring and control, which means operators are unaware of the amount of power generated, stored, or consumed.

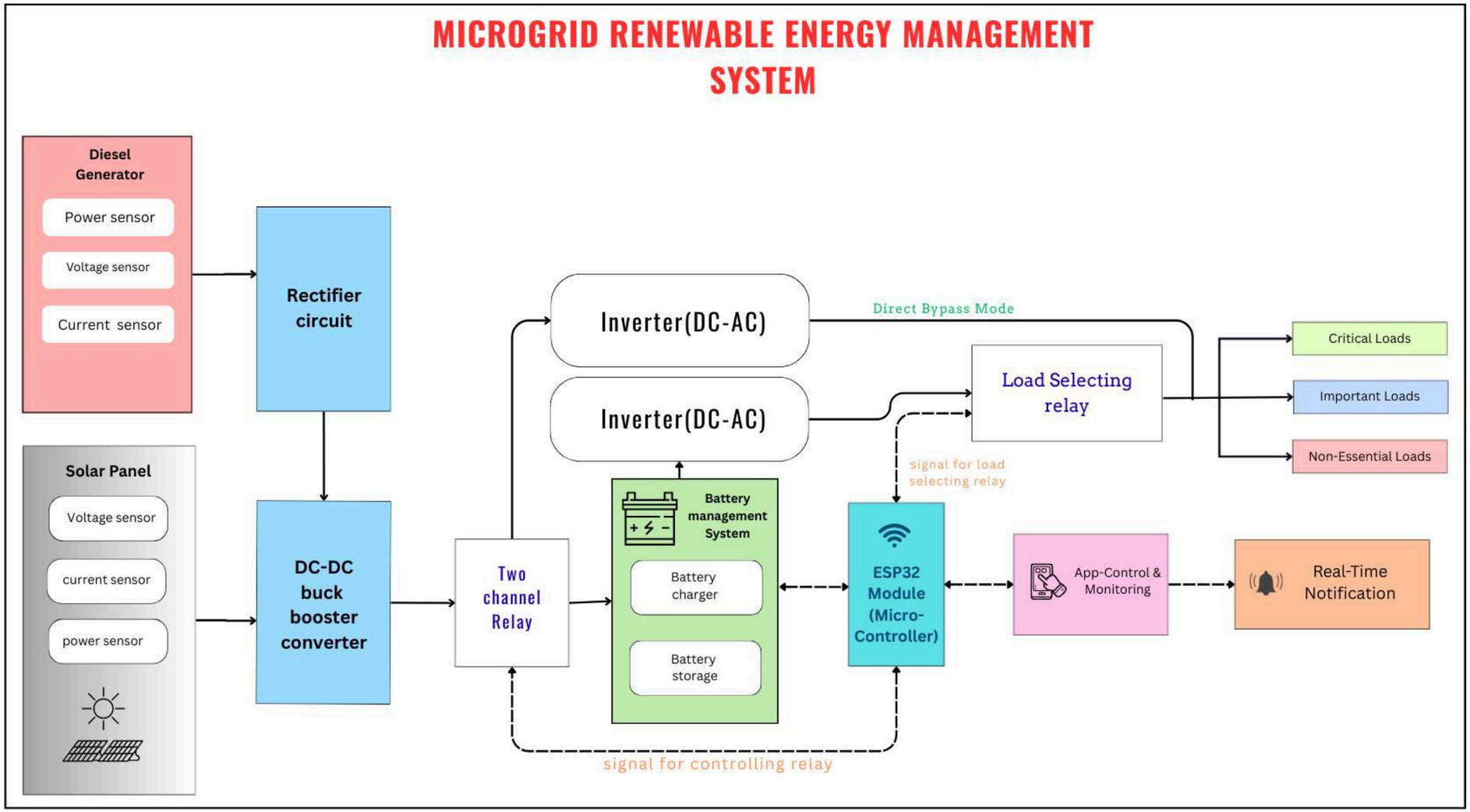
The proposed system solves this by using IoT-based monitoring and a mobile app. It shows real-time data of solar panels, battery status, and load consumption.





Uniqueness: Unlike existing solutions, this system integrates real-time monitoring, smart load prioritization, direct bypass operation, and immediate fault alerts in a single IoT platform, providing comprehensive control.

TECHNICAL APPROACH



FEASIBILITY AND VIABILITY

ECONOMIC FEASIBILITY

- Low-cost hardware: Utilizes affordable components like ESP32 microcontrollers, sensors, and Battery Management Systems (BMS), reducing upfront investment Leverages existing
- infrastructure: Compatible with current solar, battery, and diesel
- microgrids, minimizing retrofitting costs Scalable web app: Enables cost-effective deployment across multiple sites without the need for specialized equipment

TECHNICAL FEASIBILITY

- Interoperability: Designed to integrate seamlessly with hybrid energy systems (solar + battery + diesel)
- IoT-enabled monitoring: Real-time data collection and control via sensors and embedded systems
- Mobile-based interface: Simplifies user interaction and remote management.

KEY CHALLENGES

- Limited internet connectivity in remote and rural regions
- Exposuretoharshenvironmentalconditions(heat,dust,rain)may
- affect hardware reliability
- Battery degradation over time can reduce data accuracy Requires effective user onboarding and training for widespread adoption

PROPOSED SOLUTIONS

Hybrid data synchronization: supports both offline and online modes

- Durable, weather-resistant enclosures to protect hardware
- Predictive analytics to monitor faults and assess battery health
- proactively
- Intuitive mobile interface with support for multiple languages to enhance accessibility

IMPACT AND BENEFITS

KEY IMPACT

- ReliablePower Supply: Smart load prioritization ensures uninterrupted power for essential services.
- Reduced Costs: Cuts diesel usage, lowering fuel and operational expenses.
- Awareness & Transparency: Real-time app data boosts community understanding of energy use.
- Improved Quality of Life: Reliable electricity enhances education, healthcare, and livelihoods.

BENEFITS

- Economic Benefits -Empowers cost-effective energy management through affordable tech and reduced operational expenses.
- Social Benefits -Improves lives by delivering reliable electricity and accessible tools to underserved communities.
- Environmental Benefits -Supports a cleaner future by promoting renewables and reducing carbon emissions.

RESEARCH AND REFERENCES

- Hina Fathima, A. and Palanisamy, K. (2015) Optimization in Microgrids with Hybrid Energy Systems: A Review. Renewable and Sustainable Energy Reviews, 45, 431-446.

  https://doi.org /10.1016/j.rser.2015.01.059

- Fadel, A.M., Ibrahim, A.M., Tamazin, A.K., Hamdy, R.A. and Abdel-Khalik, A.S. (2019) IoT-Based Power Management for DC Microgrids. 2019 21st International Middle East Power Systems Conference (MEPCON), Cairo, 17-19 December 2019, 1107-1111.

  https://doi.org/10.1109/MEPCON47431.2019.9008216

- Hina Fathima, A. and Palanisamy, K. (2015) Optimization in Microgrids with Hybrid Energy Systems: A Review. Renewable and Sustainable Energy Reviews, 45, 431-446.

  https://doi.org/10.1016/j.rser.2015.01.059